

Ultimate Nutrition

Ultra Vitamin C-1000 Calcium Ascorbate

Antioxidants are a group of vitamins, minerals, nutrients and other compounds that the body needs to scavenge free radicals. Antioxidants work together in the body to maintain health and vitality throughout our life spans. They accomplish this by neutralizing free radicals, which can cause serious damage to the living cell, including the DNA.

The body generates free radicals in the normal course of energy production and in cases of environmental assault, such as bacterial or viral infections. Infections trigger the production of free radicals to rid the body of the invading microorganisms. Free radicals are also produced in the case of intense exercise and endurance training when the energy requirement of the body is particularly high. Thus, a certain level of free radicals is not only always present in the body but it is also essential to support normal metabolism. It is when the body cannot neutralize free radical production, which is also termed as oxidative load, that free radicals begin to exact a stiff toll on the body. In fact, free radicals are known to contribute to a variety of chronic or age-related diseases.

Antioxidants control oxidative stress and, as such, can make the difference between health and disease as well as how fast or well the body ages. A diet rich in antioxidants and intake of antioxidant supplement can ensure healthier and longer life.

The health benefits of vitamin C are abundant and varied, but it's probably best known as a cell protector, immunity booster, and powerful antioxidant. The body's ligaments, tendons, and collagen (a protein found in connective tissues) rely on the presence of vitamin C to stay strong and healthy. Like all antioxidants, vitamin C counters the effects of cell-damaging molecules called free radicals.

As an added benefit, it even helps the body recycle other antioxidants. Vitamin C helps the body to repair and maintain itself by reinforcing cell walls and helping to strengthen tiny blood vessels called capillaries. It further accelerates healing by inhibiting inflammation.

Ultimate Nutrition uses a buffered form of Vitamin C for a gentler effect on the stomach. One serving of Ultra Vitamin C-1000 contains 1000mg of premium calcium ascorbate. This product does not contain any starch, wax, preservatives, yeast, artificial color or flavor.

SELECTED REFERENCES

Balentine, J., *Pathology of Oxygen Toxicity*, New York: Academic Press, 1982

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Halliwell *et al.*, Free Radicals, Antioxidants and Human Disease: Where Are We Now?, J. Lab. Clin. Med.: 119, 598, 1996

Williams, R., Zinc: What Is Its Role in Biology, Endeavor: 8, 65, 1989

Combs, G. & Combs, S., *The Role of Selenium in Nutrition*, Orlando, Florida: Academic Press, 1986.

FREQUENTLY ASKED QUESTIONS

1 What is a Free Radical?

In the structure of atoms and molecules, electrons usually associate in pairs. When an atom or a molecule that contains one or more unpaired electron, it is classified as a free radical.

2 What Free Radicals Are Made in the Human Body?

The most commonly known free radical produced in the body is the superoxide radical. When the superoxide radical is scavenged, under certain circumstances, the body also produces hydroxyl radical. This viciously reactive radical, once generated, is practically indestructible. It triggers a chain reaction of free radical generation destroying cell membranes, proteins in the cell, and damaging DNA.

3 Do Free Radicals Delay the Aging Process?

One of the aging theories holds that free radicals induce and accelerate the aging process. Whether antioxidants delay or retard the aging process remains to be experimentally shown. Research efforts in this are being carried out in a number of laboratories.